Angle sum and difference identities

1 Consider the expression $\sin(\alpha+\beta)$.

a Distribute $\sin(\alpha+\beta)$.

b By replacing β with $-\beta$, determine the distribution of $\sin(\alpha-\beta)$.

2 Distribute $\tan(\theta-x)$.

3 Distribute and simplify $\cos(\theta-30^\circ)$.

4 Distribute $\sin(3x-y)$, rewriting it in terms of angles $3x$ and y.

5 Distribute $\cos(2x+3y)$, rewriting it in terms of angles $2x$ and $3y$.

6 Find the exact value of the following:

a $\tan 15^\circ$

b $\sin 15^\circ$

c $\cos 105^\circ$

d $\sin 105^\circ$

e $\cos 75^\circ$

f $\tan 75^\circ$

g $\cos(-15^\circ)$

h $\sin 175^\circ \cos 25^\circ - \cos 175^\circ \sin 25^\circ$

i $\dfrac{\tan 38^\circ - \tan(-22^\circ)}{1+\tan 38^\circ \tan(-22^\circ)}$

j $\sin 20^\circ \cos 40^\circ + \sin 40^\circ \cos 20^\circ$

7 Rewrite the following as a function of x:

a $\cos((180^\circ)+x)$

b $\sin(x-360^\circ)$

c Rewrite $\sin(x+90^\circ)$

d $\tan(180^\circ-x)$

e $\sin(45^\circ+x)$

f $\cos(300^\circ-x)$

g $\tan(225^\circ+x)$

h $\sin(135^\circ-x)$

8 Given that $\sin A = \dfrac{24}{25}$ and $\tan B = \dfrac{20}{21}$ where A and B are acute angles:

a Find the exact value of $\tan(A-B)$.

b Find the exact value of $\tan(A+B)$.

9 Rewrite the following as a function of x:

a $\sin\left(\frac{\pi}{[illegible]} - x\right)$

b $\cos\left(\frac{\pi}{[illegible]} + x\right)$

c $\sin\left(\frac{\pi}{[illegible]} + x\right)$

d $\tan(2\pi - x)$

10 Given that $A+B=\frac{\pi}{2}$, express $\sin\left(\frac{\pi}{2}-A\right)+\sin\left(\frac{\pi}{2}+A\right)$ as a product in terms of B.

11 Find the exact value of the following:

a $\sin\frac{5\pi}{36}\cos\frac{\pi}{36}+\sin\frac{\pi}{36}\cos\frac{5\pi}{36}$

b $\cos\frac{21\pi}{10}\cos\frac{\pi}{10}+\sin\frac{21\pi}{10}\sin\frac{\pi}{10}$

c $\dfrac{\tan\frac{2\pi}{9}-\tan\frac{\pi}{18}}{1+\tan\frac{2\pi}{9}\tan\frac{\pi}{18}}$

12 Suppose $\cos A=\frac{3}{5}$ and $\sin B=\frac{15}{17}$ where $0<A<\frac{\pi}{2}$ and $0<B<\frac{\pi}{2}$:

a Find $\sin A$.

b Find $\cos B$.

c Find $\sin(A-B)$.

d Find $\cos(A-B)$.

e In which quadrant is angle $(A-B)$?

f Find $\tan(A-B)$.

13 Suppose $\sin A=\frac{3}{5}$ and $\sin B=-\frac{12}{13}$ where $0<A<\frac{\pi}{2}$ and $\pi<B<\frac{3\pi}{2}$:

a Find $\cos A$.

b Find $\cos B$.

c Find $\sin(A+B)$.

d Find $\sin(A-B)$.

e Find $\tan(A+B)$.

f Find $\tan(A-B)$.

g In which quadrant is $A+B$?

h In which quadrant is $A-B$?

14 For each of the following:

i Write the angle as a sum or difference of two angles.

ii Find the exact value of the ratio.

a $\cos\frac{7\pi}{12}$

b $\sin\frac{11\pi}{12}$

c $\tan\frac{\pi}{12}$

d $\tan\left(-\frac{\pi}{12}\right)$

15 Express the following in simplest form:

a $\sin A\cos 2B+\cos A\sin 2B$

b $\cos(3\theta+x)\cos 3\theta-\sin(3\theta+x)\sin 3\theta$

c $\dfrac{\tan 4m-\tan m}{1+\tan 4m\tan m}$

d $\sin(\pi+\theta)-\sin(\pi-\theta)$

16 Express each of the following as a product o[illegible] trigonometric functions:

a $\sin(6x) + \sin(4x)$

b $\cos 3x - \cos 5x$

c $\cos\left(\dfrac{3x+y}{2}\right) + \cos\left(\dfrac{x+3y}{2}\right)$

d $\sin 44^\circ + \sin 18^\circ$

17 Evaluate $\cos 195^\circ - \cos 75^\circ$ by first expressing it as a product of two trigonometric functions.

18 Solve the following equation for x:

$$x^2 - \sqrt{2}\sin(45^\circ + \theta)\,x + \sin\theta\cos\theta = 0$$

19 Solve the following equation for $0^\circ \leq x \leq 360^\circ$:

$$\sin 4x \cos x - \sin x \cos 4x = \frac{\sqrt{3}}{2}$$

Proofs

20 Verify that $\cos(x + 210^\circ) + \sin(x + 120^\circ) = 0$.

21 Prove the following identities:

a $\sin(x+y) - \sin(x-y) = 2\cos x \sin y$

b $\dfrac{\sin(x-y)}{\cos x \cos y} = \tan x - \tan y$

c $\dfrac{\cos(x+y)}{\cos y} - \dfrac{\sin(x+y)}{\sin y} = -\dfrac{\sin x}{\sin y \cos y}$

d $\dfrac{\tan(x-y) + \tan y}{1 - \tan(x-y)\tan y} = \tan x$

e $\tan(\theta+\alpha)\tan(\theta-\alpha) = \dfrac{\tan^2(\theta) - \tan^2(\alpha)}{1 - \tan^2(\theta)\tan^2(\alpha)}$

f $\dfrac{\tan A - \tan B}{\tan A + \tan B} = \dfrac{\sin(A-B)}{\sin(A+B)}$

g $\dfrac{1 + \tan x \tan y}{\tan x - \tan y} = \cot(x-y)$

h $\cot(x+y) = \dfrac{\cot x \cot y - 1}{\cot x + \cot y}$

22 Consider $2x = x + x$, and verify the followin

a $\cos 2x = 2\cos^2(x) - 1$

b $\sin 2x = 2\sin x \cos x$

23 If $\tan A + \tan B = -5$ and $\tan A \tan B = 6$, prove that $\sin(A+B) = \cos(A+B)$.

Applications

24 The pendulum in a grandfather clock is pulled to its maximum displacement on one side and then released. The displacement, in centimeters, of the pendulum from its vertical equilibrium position is given by the formula:

$$D = 17\sin\left(\pi\left(x + \frac{1}{2}\right)\right)$$

where x is the time in seconds since the pendulum was released.

a Find the maximum displacement of the pendulum.

b Rewrite the formula for D in terms of the cosine function.

c Find the period of oscillation of the pendulum. That is, how long does the pendulum take to return to the maximum displacement from where it started?

25 The average depth of water, in meters, of a certain harbor can be modeled by the function:

$$h = \frac{1}{2}\sin\left(\frac{\pi}{6}x + \frac{2\pi}{3}\right) + \frac{5}{2}$$

where x is the time in hours since midnight. The average depth of water has been graphed over a 24-hour period.

a Using the graph, determine the best estimate of the average depth of the water at 3 am, correct to one decimal place.

b Calculate the average depth of the water in the harbour at 5:00 pm.

c Rewrite the function in Distributeed form, in terms of $\sin\left(\frac{\pi}{6}x\right)$ and $\cos\left(\frac{\pi}{6}x\right)$
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